

Uses



This bulb contains yttrium compounds that enable it to produce a warmer, more yellow glow.

LED lamp



Laser

This laser is powered using crystals composed of yttrium and silicon, and can cut through metal.



Yttrium-90

This radioactive form of yttrium is used to treat cancers in the body.

Shock-proof lenses are made from yttrium-infused glass to make them tough.

Yttrium gas mantle

Digital camera lens

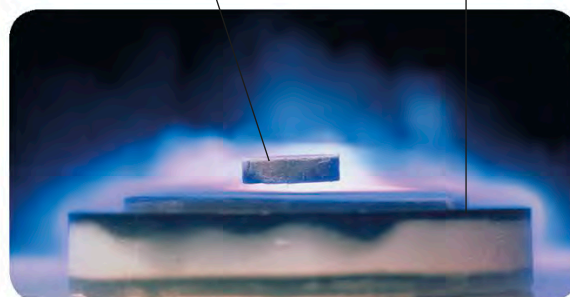


This fabric mantle holds the hot flame inside.



A small magnet floats over the superconductor.

This superconductor produces a magnetic field that repels the magnet above it.



Yttrium superconductor

NASA spacecraft use yttrium lasers to map the **surface of asteroids** in space.

LED lamps, yttrium converts blue light to other colours. Many **lasers** use an artificial mixture of yttrium and aluminium inside a silicon-rich crystal called garnet. Powerful yttrium lasers are used for treating some skin infections, as well as by dentists during tooth surgery. A radioactive

form of this element has medical applications. Yttrium is added to the glass in a **camera lens** to make it tough. Yttrium compounds are also used in **superconductors** – materials that conduct electricity easily when cooled to very low temperatures.



FRIEDRICH WÖHLER

In 1828, the German chemist Friedrich Wöhler became the first person to purify yttrium. He did so by extracting it from the compound yttrium chloride. He was also the first person to extract the metals beryllium and titanium from their ores.

